

# Logistic Regression - Data Analysis and Visualization

Dr. Farshid Alizadeh-Shabdiz  
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# Classification

# Classification

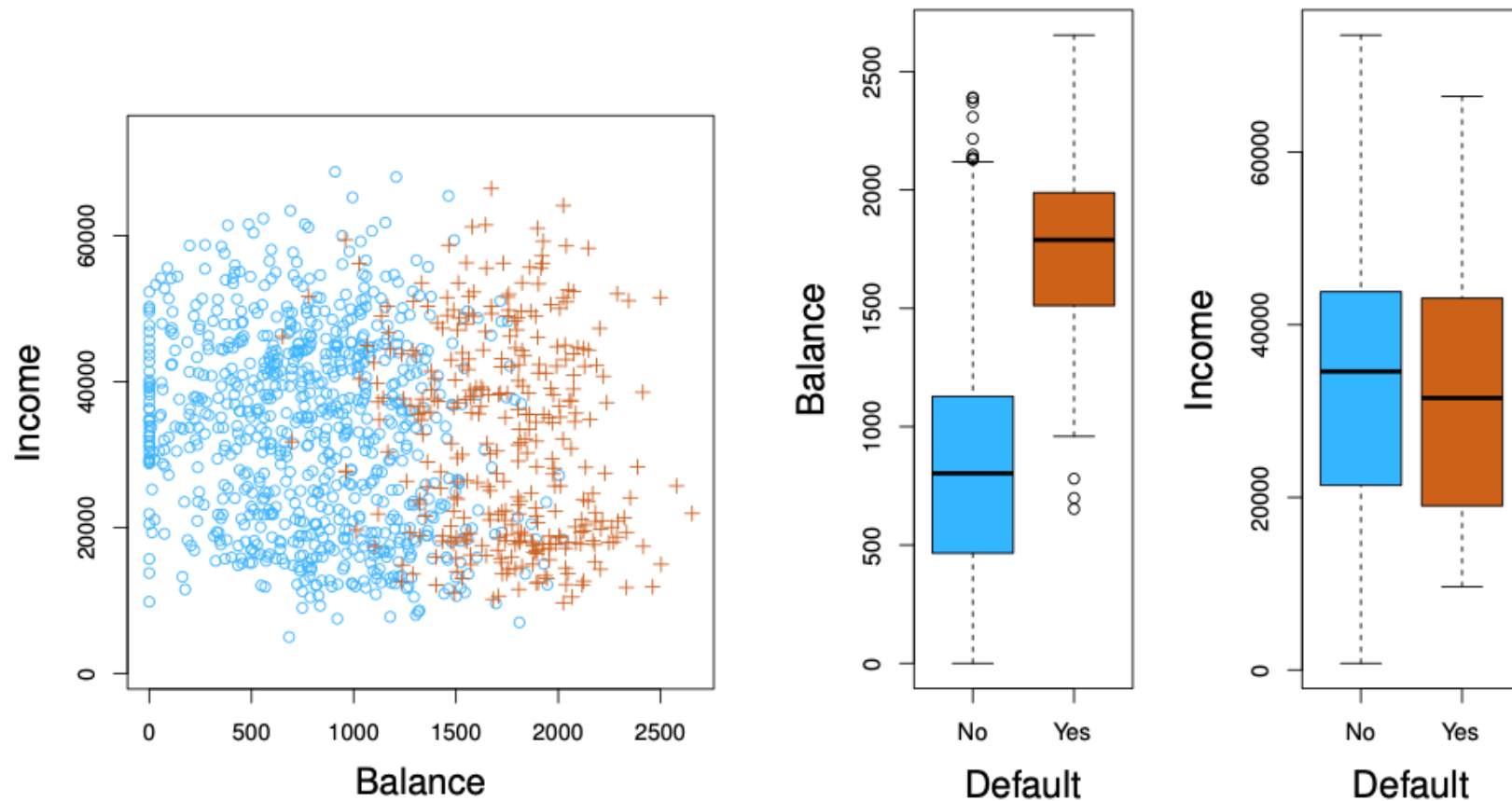
## Classification is

- Predicting a qualitative output from
- A set of quantitative inputs/parameters,  $X$
- And most of the time, also estimating quality of the estimate

## For example

- Detecting spam email
- Detecting credit card fraud
- Diagnosing a patient's illness
- Object detection
- Face recognition
- News article classification
- Recommending products to customers

# Example of Classification



Ref: In-depth introduction to machine learning, by T. Hastie and R. Tibshirani

# Linear Regression as classification

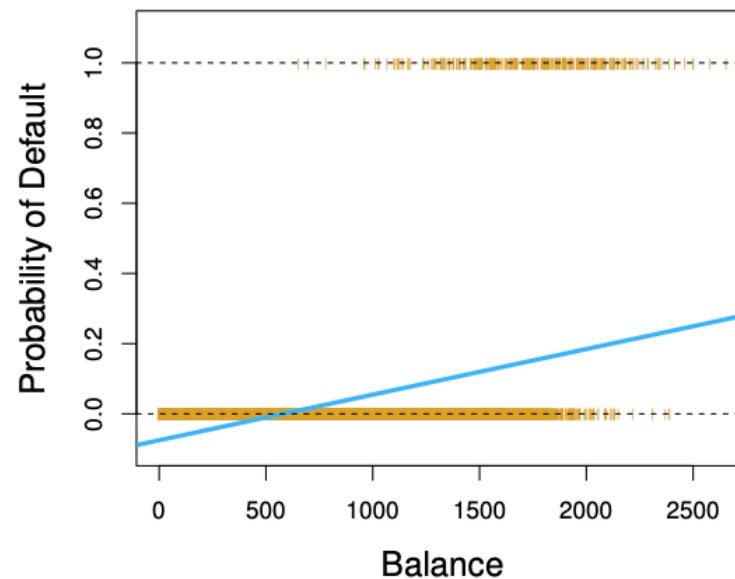
- Can linear regression solve classification with output

$$y = \begin{cases} 0 \\ 1 \end{cases} \text{ \& Threshold = 0.5}$$

- Short answer is YES – linear regression does a good job,  
**BUT**

- Linear regression can produce an output outside  $[0,1]$  !!
- It can give a probability much higher than one for class-1
- Also a probability much lower than zero for class-0
- Many assumption of linear regression are not met

# Linear Regression



Ref: In-depth introduction to machine learning, by T. Hastie and R. Tibshirani

# Why Logistic Function

- The goal is
  - Finding probability of a success event, which is  $p$  (Diagnose is C).
  - Using linear combination of parameters to estimate  $p$
- Note  $p$  changes between  $[0,1]$ , but linear combination of parameters change between  $(-\infty, \infty)$  – Big discrepancy
- So probability of no-success is  $(1-p)$
- The “odds ratio” is defined as  $(\frac{p}{1-p})$ , which mean how odd is having a success
- The odds ratio varies between  $[0, \infty)$ , closer to linear output
- Easy way to map a range of real positive to real numbers is *log* function
- So, we have it!

# Logistic Regression

➤ So the output  $y$  will be found as

$$y = \frac{e^{\beta_0 + \beta_1 X_1}}{1 + e^{\beta_0 + \beta_1 X_1}}$$

- $e$  ( $\sim 2.71828$ ) is constant. It is called Euler's number or "natural number"
- Note ( $0 \leq y \leq 1$ )

Another way of looking at logistic Function

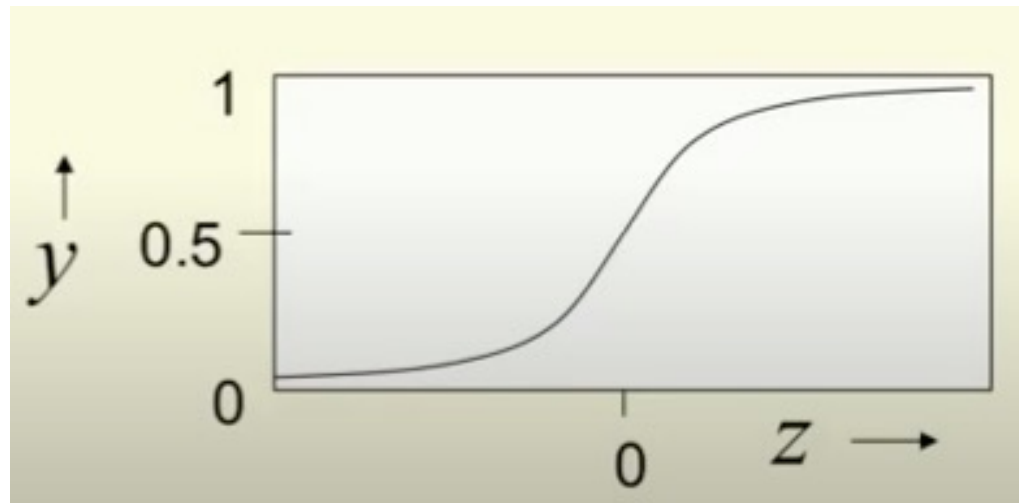
$$Z = \beta_0 + \beta_1 X_1$$
$$y = \frac{e^Z}{1 + e^Z}$$



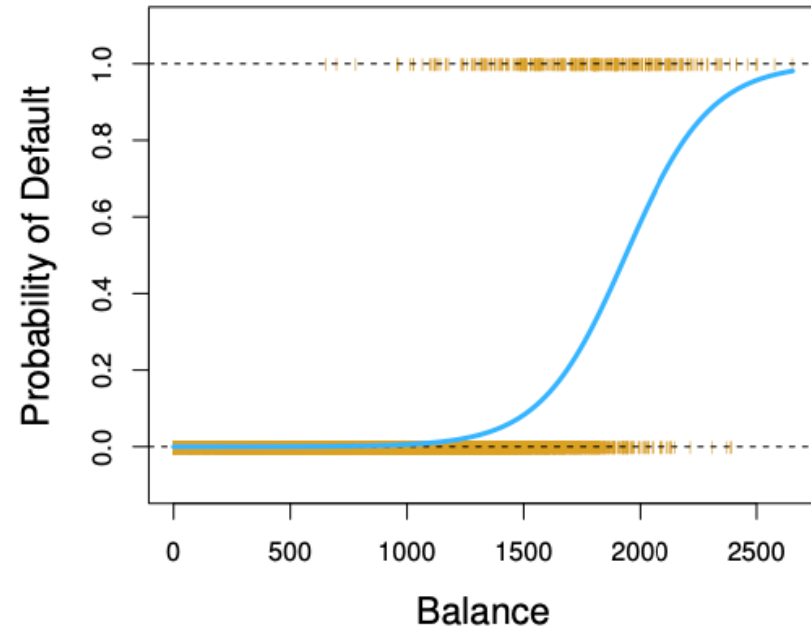
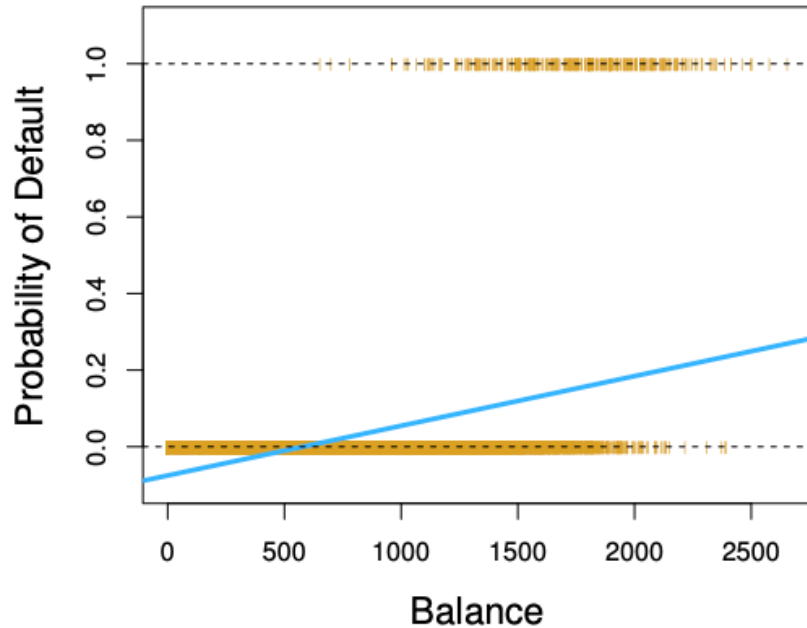
# Logistic Function Is a Good Fit

- Logistic Function:

$$y = \frac{e^z}{1+e^z}$$



# Linear vs Logistic Regression



Ref: In-depth introduction to machine learning, by T. Hastie and R. Tibshirani

- The above example is a yes and no answers. Orange dots are marking the answers.

# Example – Prediction of Default

- Predicting default as a function of balance

```
> glm(default ~ balance , data=Default , family=binomial)

Call:  glm(formula = default ~ balance, family = binomial, data = Default)

Coefficients:
(Intercept)      balance 
 -10.651331      0.005499 

Degrees of Freedom: 9999 Total (i.e. Null);  9998 Residual
Null Deviance:      2921 
Residual Deviance: 1596      AIC: 1600
```

|                  | Coefficient | Std. Error | Z-statistic | P-value  |
|------------------|-------------|------------|-------------|----------|
| <b>Intercept</b> | -10.6513    | 0.3612     | -29.5       | < 0.0001 |
| <b>balance</b>   | 0.0055      | 0.0002     | 24.9        | < 0.0001 |

## Example – Prediction of Default Examples

- Probability of a person with \$1000 balance default

$$y = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} = \frac{e^{-10.6 + 0.0055 \times 1000}}{1 + e^{-10.6 + 0.0055 \times 1000}} = 0.006$$

What if balance is \$2000

$$y = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} = \frac{e^{-10.6 + 0.0055 \times 2000}}{1 + e^{-10.6 + 0.0055 \times 2000}} = 0.586$$

# Multivariable Logistic Regression

➤ So the output  $y$  will be found as

$$y = \frac{\exp(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}{1 + \exp(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}$$

Note that ( $0 \leq y \leq 1$ )

And

$$y = \begin{cases} 0, & \text{Default} \\ 1, & \text{No default} \end{cases}$$

# Example –Default for Students

- What about probability of default for students

```
> glm(default ~ student , data=Default , family=binomial)
```

```
Call: glm(formula = default ~ student, family = binomial, data = Default)
```

```
Coefficients:
```

```
(Intercept)    studentYes  
    -3.5041         0.4049
```

```
Degrees of Freedom: 9999 Total (i.e. Null); 9998 Residual
```

```
Null Deviance:      2921
```

```
Residual Deviance: 2909      AIC: 2913
```

|                      | Coefficient | Std. Error | Z-statistic | P-value  |
|----------------------|-------------|------------|-------------|----------|
| <b>Intercept</b>     | -3.5041     | 0.0707     | -49.55      | < 0.0001 |
| <b>student [Yes]</b> | 0.4049      | 0.1150     | 3.52        | 0.0004   |

- Predicting default as a function of “Student” being an student.

$$\widehat{\text{Pr}}(\text{default}=\text{Yes}|\text{student}=\text{Yes}) = \frac{e^{-3.5041+0.4049 \times 1}}{1 + e^{-3.5041+0.4049 \times 1}} = 0.0431,$$

$$\widehat{\text{Pr}}(\text{default}=\text{Yes}|\text{student}=\text{No}) = \frac{e^{-3.5041+0.4049 \times 0}}{1 + e^{-3.5041+0.4049 \times 0}} = 0.0292.$$

# Example – Default vs Several Variables

- Predicting default as a function of “Student”, income, and balance is as follows.

```
> glm(default ~ . , data=Default , family=binomial)

Call:  glm(formula = default ~ ., family = binomial, data = Default)

Coefficients:
(Intercept)  studentYes      balance      income
-1.087e+01  -6.468e-01   5.737e-03   3.033e-06

Degrees of Freedom: 9999 Total (i.e. Null); 9996 Residual
Null Deviance:      2921
Residual Deviance: 1572      AIC: 1580
```

|                      | Coefficient | Std. Error | Z-statistic | P-value  |
|----------------------|-------------|------------|-------------|----------|
| <b>Intercept</b>     | -10.8690    | 0.4923     | -22.08      | < 0.0001 |
| <b>balance</b>       | 0.0057      | 0.0002     | 24.74       | < 0.0001 |
| <b>income</b>        | 0.0030      | 0.0082     | 0.37        | 0.7115   |
| <b>student [Yes]</b> | -0.6468     | 0.2362     | -2.74       | 0.0062   |



# Multivariable Logistic Regression

- Why student became negative here ?!

Note: correlation between variables in multivariable logistic regression can make inference hard.

```
> glm(default ~ . , data=Default , family=binomial)
```

```
Call:  glm(formula = default ~ ., family = binomial, data = Default)
```

```
Coefficients:
```

| (Intercept) | studentYes | balance   | income    |
|-------------|------------|-----------|-----------|
| -1.087e+01  | -6.468e-01 | 5.737e-03 | 3.033e-06 |

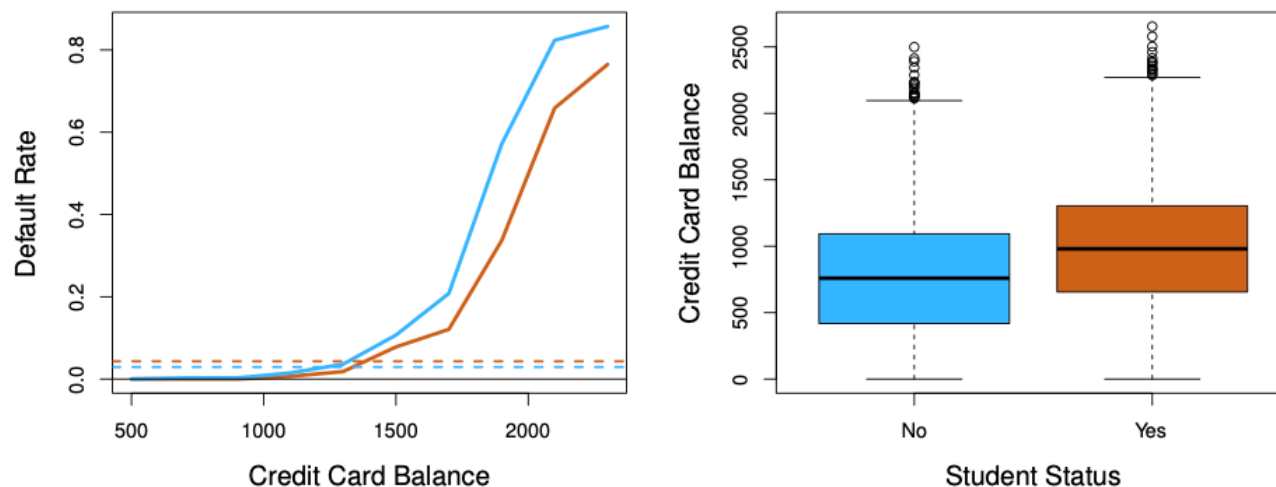
```
Degrees of Freedom: 9999 Total (i.e. Null); 9996 Residual
```

```
Null Deviance: 2921
```

```
Residual Deviance: 1572 AIC: 1580
```

# Logic of the Results for Students

- Students have higher balance, so it is more likely for students to default
- But for a given balance, students default is lower



Ref: In-depth introduction to machine learning, by T. Hastie and R. Tibshirani

# Interesting Math about Logistic Regression

- Logistic function has interesting mathematical properties

1.  $\ln \left( \frac{y}{1-y} \right) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$

- This is called *log odds* or *logit* transformation of  $y$

2.  $\frac{\partial y}{\partial z} = y(1 - y)$  - derivative is important for numerical solutions

# Odd ratio of Logistic Regression

$$\frac{y}{1-y} = \exp(\beta_0 + \beta_1 X_1 + \cdots + \beta_n X_n)$$

## Interpretation

$$\hat{p} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x}}$$

- ▷ The regression equation can be used to predict risk of the event.
- ▷ The interpretation of the regression coefficient(s) are generally based on odds ratios.

**Consider the odds ratio of an event for a given value of  $x = x_a$  versus a given value of  $x = x_b$ .**

- ▷ The estimated odds for a given value of  $x = x_a$  is given by  $\widehat{\text{odds}}_a = e^{\hat{\beta}_0 + \hat{\beta}_1 x_a}$
- ▷ The estimated odds for a given value of  $x = x_b$  is given by  $\widehat{\text{odds}}_b = e^{\hat{\beta}_0 + \hat{\beta}_1 x_b}$

The odds ratio then is given by

$$\widehat{OR}_{x_a \text{ versus } x_b} = \frac{\widehat{\text{odds}}_a}{\widehat{\text{odds}}_b} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x_a}}{e^{\hat{\beta}_0 + \hat{\beta}_1 x_b}}$$

## Interpretation

The odds of the event are  $e^{\hat{\beta}_1(x_a - x_b)}$  higher for every  $x_a - x_b$  unit increase in  $x$ .

**Interpretation depends only on the difference in  $x$  values as opposed to their actual values.**

$$\begin{aligned}\widehat{OR}_{x_a \text{ versus } x_b} &= \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x_a}}{e^{\hat{\beta}_0 + \hat{\beta}_1 x_b}} \\ &= e^{(\hat{\beta}_0 + \hat{\beta}_1 x_a) - (\hat{\beta}_0 + \hat{\beta}_1 x_b)} \\ &= e^{\hat{\beta}_0 + \hat{\beta}_1 x_a - \hat{\beta}_0 - \hat{\beta}_1 x_b} \\ &= e^{\hat{\beta}_1 x_a - \hat{\beta}_1 x_b} \\ &= e^{\hat{\beta}_1 (x_a - x_b)}\end{aligned}$$

## Confidence Interval

Confidence intervals for the logistic regression setting are based on the odds ratio.

The two-sided  $100\% \times (1 - \alpha)$  confidence interval for  $\widehat{OR}_{x_a \text{ versus } x_b}$  is:

$$e^{\left(\hat{\beta}_1 \pm z_{\frac{\alpha}{2}} \cdot SE_{\hat{\beta}_1}\right)(x_a - x_b)}$$

Where

- ▷  $SE_{\hat{\beta}_1}$  is the standard error of the regression coefficient and
- ▷  $z_{\frac{\alpha}{2}}$  is the value from the standard normal distribution with a right tail probability of  $\alpha/2$ .

## An Example: Logistic Regression

We are interested in the association between cholesterol levels and having a coronary event in a high-risk patient population (who have had an event in the past). We collect cholesterol data for 50 subjects and then follow each for a year to see if they have another coronary event.

**Explanatory variable is cholesterol level and our outcome is whether or not the subject had another coronary event.**

Given the nature of our response variable, we perform a logistic regression. A summary of the beta estimates from the model are shown below.

| Parameter | Estimate | Standard Error | p-value   |
|-----------|----------|----------------|-----------|
| $\beta_0$ | -3.3848  | 0.5838         | < 0.00001 |
| $\beta_1$ | 0.1253   | 0.0362         | 0.0005    |



## An Example: Logistic Regression

| Parameter | Estimate | Standard Error | p-value |
|-----------|----------|----------------|---------|
| $\beta_0$ | -3.725   | 1.753          | 0.0336  |
| $\beta_1$ | 0.024    | 0.012          | 0.0420  |

Use these results to **how is the SE is calculated?**

- ▷ **predict the risk of another coronary event** for a high risk patient with a cholesterol level of 190.
- ▷ **calculate the odds ratio** for a coronary event of a high-risk patient with a cholesterol level of 190 versus a patient with a cholesterol level of 180.
- ▷ **calculate 95% confidence interval** for the odds ratio of having a coronary event for a patient with a cholesterol level of 190 versus a patient with a cholesterol level of 180.

## An Example: Logistic Regression

| Parameter | Estimate | Standard Error | p-value |
|-----------|----------|----------------|---------|
| $\beta_0$ | -3.725   | 1.753          | 0.0336  |
| $\beta_1$ | 0.024    | 0.012          | 0.0420  |

The risk of having a coronary event for a patient with a cholesterol level of 190 is predicted by :

$$\hat{p} = \frac{e^{-3.725+0.024*190}}{1 + e^{-3.725+0.024*190}} = \frac{e^{0.835}}{1 + e^{0.835}} = 0.697$$

## An Example: Logistic Regression

The odds ratio of having a coronary event for a patient with a cholesterol level of 190 versus a patient with a cholesterol level of 180 is

$$\hat{OR}_{X_a \text{ versus } X_b} = e^{\hat{\beta}_1 (X_a - X_b)} = e^{0.024 * (190 - 180)} = e^{0.24} = 1.27$$

### Interpretations:

- ▶ *The odds of having a coronary event are 1.27 times higher for every 10 Unit increase in cholesterol level.*
- ▶ *The odds ratio comparing any two individuals with cholesterol levels which are 10 units apart are the same.*

**The quantity  $e^{\hat{\beta}_1}$  is the odd ratio of the event for two individuals with x values that are 1 unit apart.**

**In other words,  $e^{\hat{\beta}_1}$  is the relative increase in odds for every 1 unit increase in x.**

## An Example: Logistic Regression

The 95% confidence interval for the odds ratio of having a coronary event for a patient with a cholesterol level of 190 versus a patient with a cholesterol level of 180 is:

$$e^{\left(\hat{\beta}_1 \pm z_{\frac{\alpha}{2}} \cdot SE_{\hat{\beta}_1}\right)(x_a - x_b)} = e^{(0.024 \pm 1.96 \cdot 0.0122) \cdot 10} \\ = (1.004, 1.608)$$

*We are 95% confident that the odds of having a coronary event are between 1.004 and 1.608 times higher for every 10-unit increase in cholesterol level.*

# Logistic Regression & Case Control Sampling

- When Logistic regression is used for rare events, the model will be trained for a none proportional ratio of samples
  - Like training with a set with 40% of rare event sample
- As a result the model will calculate probabilities wrong!

## Solution:

- Case control sampling
  - Regression parameters  $\beta_i$  are accurate, and only the intercept  $\beta_0$  is not, which gets corrected by

$$\beta_0^* = \beta_0 + \log\left(\frac{p_{rare}}{1 - p_{rare}}\right) - \log\left(\frac{p_{set}}{1 - p_{set}}\right)$$

- $P_{rare}$ : actual probability of the rare event how come?
- $P_{set}$ : probability of the rare event in the training set

# Control vs Case Sample Size

- Control to case ratio: In order to have smaller variance in the coefficients it is good to have more control samples. what?
- Question is how much more?
  - Rule of thumb is five to six times is sufficient

# What about Multi Classification

- For example
  - Classifying news articles to sport, politics, family, kids, etc.
  - Classifying different people in a picture

# Multiclass Logistic Regression or Multinomial Regression

- Logistic regression can be easily extended to more than two class prediction.

$$\Pr(y = k|X) = \frac{e^{\beta_{0k} + \beta_{1k}X_1 + \dots + \beta_{nk}X_n}}{\sum_{j=1}^K e^{\beta_{0j} + \beta_{1j}X_1 + \dots + \beta_{nj}X_n}}$$

K : capital K is total number of classes

k: small K is one of the classes

- Select the class with the highest probability
- This is also called “softmax” function
- Note - Linear regression cannot solve this problem



## Generalized linear models (GLMs)<sup>2</sup> what?

- ▷ We can use ANY stochastic process for generating the error, not just normal distribution.
- ▷ GLM is an extension of linear regression that allows errors to be generated by a wide variety of distributions.
- ▷ In particular, any distributions in the “Exponential Family”
- ▷ GLMs extend the linear modeling capability of R to scenarios that involve non-normal error distributions. The idea is to obtain linear functions of the predictor variables by transforming the right side of the equation by a link function.

| Error Family | Link     | Inverse of link     | Used for           |
|--------------|----------|---------------------|--------------------|
| Gaussian     | identity | 1                   | normally error     |
| Poisson      | log      | exp                 | counts             |
| Binomial     | logit    | $1/(1 + 1/\exp(x))$ | proportions        |
| Gamma        | inverse  | $1/x$               | non-constant error |

<sup>2</sup>[https://en.wikipedia.org/wiki/Generalized\\_linear\\_model](https://en.wikipedia.org/wiki/Generalized_linear_model)

# Logistic Regression Function in R

- Logistic regression function in R provides
  - Coefficients of the model
  - p-value of the parameters same as linear regression
  - Also, “Z-statistics”, which is coefficient for normalized parameters

how is standard error calculated?  
what is z\_value?

```
> tmp = glm(default ~ . , data=Default , family=binomial)
> summary(tmp)
```

Call:  
glm(formula = default ~ . , family = binomial, data = Default)

Deviance Residuals:

| Min     | 1Q      | Median  | 3Q      | Max    |
|---------|---------|---------|---------|--------|
| -2.4691 | -0.1418 | -0.0557 | -0.0203 | 3.7383 |

Coefficients:

|             | Estimate   | Std. Error | z value | Pr(> z )    |
|-------------|------------|------------|---------|-------------|
| (Intercept) | -1.087e+01 | 4.923e-01  | -22.080 | < 2e-16 *** |
| studentYes  | -6.468e-01 | 2.363e-01  | -2.738  | 0.00619 **  |
| balance     | 5.737e-03  | 2.319e-04  | 24.738  | < 2e-16 *** |
| income      | 3.033e-06  | 8.203e-06  | 0.370   | 0.71152     |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## R Commands: Generalized Linear Models (GLMs)

Use the `glm()` function with binomial option

```
> glm(data$event ~ data$explanatory1 + data$explanatory2 + ... , family=
      binomial)
```

- ▷ Ensure that your event is coded as 1 = *Event* and 0 = *non – event* (numeric, not a factor variable)
- ▷ If one of the variables in the model is a factor variable, it is best to create dummy variables (1/0) so that you know exactly what the reference group is
- ▷ “**family**” parameter is a simple way of specifying a choice of variance and link functions When family is set to binomial, it tells R to perform logistic regression. **what?**

<http://plantecology.syr.edu/fridley/bio793/glm.html>

## R commands: Generalized Linear Models (GLMs)

```
> glm(data$event~data$explanatory1 + data$explanatory2 + ... , family=
      binomial)
```

- ▷ Use the **summary()** function on the saved regression result to get regression equation and associated tests for each regression coefficient
- ▷ Use the **exp()** function, which computes the exponential value of a number  $e^x$ , on the resulting coefficients to obtain odds ratios for each regression coefficient
- ▷ Use the **predict()** function on the saved regression result to get the predicted risks for each observations

## Logistic Regression: R commands

```
> data <- read.csv('cevent.csv')
# Simple logistic regression
> m <- glm(data$event ~ data$chol, family=binomial)
> summary(m)

Call:
glm(formula = data$event ~ data$chol, family = binomial)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.5752   -0.9629   -0.7217    1.1418    2.1732

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.72518    1.75307  -2.125   0.0336 *
data$chol    0.02359    0.01160   2.034   0.0420 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

---
```

# Logistic Regression In R

- `install.packages("ISLR")`
- `library(ISLR)`
- Function
  - `glm(y ~ X, data=DataName ,family = binomial)`
  - `glm(y ~ X+Z, data=DataName ,family = binomial)`
  - `glm(y ~ ., data=DataName ,family = binomial)`
- Load data “Default”
  - Default as a function of Balance
  - Default as a function of Student
    - `Default$studentBinFlag = ifelse(student=="Yes", 1, 0)`
  - Default as a function of everything

Important note:

Logistic Regression is not Stable for fully separable classes. why?

In this case, other classification methods like Discriminant analysis has to be used.

# Formal Inference in Logistic Regression

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# Formal Inference in Simple Logistic Regression

Formal inference in Simple Logistic Regression uses estimates of  $\hat{\beta}_1$ .

$H_0 : \beta_1 = 0$  ( $H_0$  : there is no association between  $x$  and odds of the outcome)

$H_1 : \beta_1 \neq 0$  ( $H_1$  : there is an association between  $x$  and odds of the outcome)

$\beta_1 = 0$  is equivalent to that the regression line had a slope of 0 (it would be a horizontal line),  $\beta_1 = 0$  means  $OR = e^{\beta_1} = 1$ .

- ▷ The null hypothesis  $\beta_1 = 0$  is equivalent to the test of the odds ratio for a 1 unit increase in  $x$  being equal to 1 ( $H_0 : OR = 1$ ).
- ▷  $H_0 : \beta_1 = 0$  or  $OR = 1$  is rejected if  $\hat{\beta}_1$  is sufficiently far from 0.
- ▷ We reject the claim that the population parameter  $\beta_1$  is equal to 0 if  $\hat{\beta}_1$ , the sample statistic, is far from 0.

## An Example: logistic regression inference

Formally test whether or not cholesterol is associated with risk of a coronary event at the  $\alpha = 0.05$  level.

**1. Set up the hypotheses and select the alpha level**  $H_0 : \beta_1 = 0$  or  $OR = 1$  (there is no association between cholesterol levels and risk for a coronary event)

$H_1 : \beta_1 \neq 0$  or  $OR \neq 1$  (there is an association between cholesterol levels and risk for a coronary event)  $\alpha = 0.05$

**2. Select the appropriate test statistic**

$$z = \frac{\beta_1}{SE_{\beta_1}}$$

**3. State the decision rule**

Determine the appropriate value from the standard normal distribution associated with a right hand tail probability of  $\alpha/2 = 0.05/2 = 0.025$

$$z_{\frac{\alpha}{2}} = 1.960$$

Decision Rule: Reject  $H_0$  if  $|z| \geq 1.96$  or Reject  $H_0$  if  $p \leq \alpha$

Otherwise, do not reject  $H_0$

## An Example: Logistic Regression Inference

### 4. Compute the test statistic

$$z = \frac{\beta_1}{SE_{\beta_1}} = \frac{0.024}{0.0116} = 2.069$$

### 5. Conclusion

Reject  $H_0$  since  $z \geq 1.96$  or since  $p\text{-value} \leq \alpha$ . We have significant evidence at the  $\alpha = 0.05$  level that  $\beta_1 \neq 0$ . There is evidence of an association between cholesterol level and risk of a coronary event.

**The odds ratio for a coronary event is  $e^{\beta_1} = 1.02$  for every 1 unit increase in cholesterol.** (Or we could say that the odds ratio is 1.27 for every 10-unit increase in cholesterol as this may be a more reasonable and clinically relevant scale to report the results).

**We are 95% confident that the true odds ratio is between 1.00 and 1.047.** (We could also report the 95% confidence interval for the 10-unit increase instead if we had chosen to present the odds ratio in the previous sentence based on this unit of increase).

## R commands: Predict Method for GLM Fits

```
# predicted risk for each patient
risk <- predict(m, type=c("response"))
risk
  1          2          3          4          5          6          7
0.22720323 0.43615030 0.34653363 0.31520904 0.23137263 0.48299515
      0.59393290
      8          9         10         11         12         13
      0.44196119 0.49478598 0.47122323 0.70593666 0.61088437 0.41309669
      0.18133869
...
```

**The parameter "type" indicates the type of prediction required.**

The default is on the scale of the linear predictors; the alternative "response" is on the scale of the response variable. **Thus for a default binomial model the default predictions are of log-odds (probabilities on logit scale) and type = "response" gives the predicted probabilities.**

## R commands: Predict Method for GLM Fits

```
> risk <- predict(m, type=c("response"))

# predicted risk for patient with cholesterol of 190
> risk[41]
41
0.6808668

# Or manual calculation
> exp(m$coefficients[1]+m$coefficients[2]*190)/(1+exp(m$coefficients[1]+
  m$coefficients[2]*190))
(Intercept)
0.6808668
```

## An Example: Multiple Logistic Regression

Our explanatory variables are cholesterol level, age and gender and our outcome is whether or not the subject had another coronary event.

**The p-value for the global test was 0.0058.**

A summary of the beta estimates from the model are shown below. Test the global null hypothesis at the  $\alpha = 0.05$  level.

| Parameter                  | Estimate | Standard Error | p-value |
|----------------------------|----------|----------------|---------|
| (Intercept)                | -8.536   | 2.684          | 0.001   |
| $\beta_{Age}$              | 0.042    | 0.025          | 0.096   |
| $\beta_{CholesterolLevel}$ | 0.029    | 0.013          | 0.024   |
| $\beta_{Gender}$           | 2.521    | 0.803          | 0.002   |

## An Example: Logistic Regression Inference

The test for the global null hypothesis tests:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$$H_1 : \text{at least one } \beta_i \neq 0$$

The p-value for the global test was 0.0058. Since , we reject the null hypothesis and conclude that there is at least one  $\beta_i \neq 0$ .

### Example Regression Coefficient for Age:

$H_0 : \beta_{age} = 0$  or  $OR_{age} = 1$  (there is no association between age and risk for a coronary event, after controlling for cholesterol level and gender)

$H_1 : \beta_{age} \neq 0$  or  $OR_{age} \neq 1$  (there is an association between age and risk for a coronary event, after controlling for cholesterol level and gender)

We fail to reject the null hypothesis that or **after adjusting for cholesterol level and gender** since  $p > \alpha$ . We do not have significant evidence at the  $\alpha = 0.05$  level that  $\beta_{age} \neq 0$  ( $p = 0.096$ ).

**The odds ratio for a coronary event is 1.04 for every 1-year increase in age.**



## An example: logistic regression inference

### Cholesterol Level:

Reject  $H_0 : \beta_{chol} = 0$  or  $OR_{chol} = 1$  after adjusting for age and gender since  $p \leq \alpha$

We have significant evidence at the  $\alpha = 0.05$  level that  $\beta_{chol} \neq 0$ .

*There is evidence of an association between cholesterol level and risk of a coronary event after adjusting for age and gender.*

**The odds ratio for a coronary event is 1.029 for every 1-unit increase in cholesterol.**

### Gender:

Reject  $H_0 : \beta_{gender} = 0$  or  $OR_{gender} = 1$  after adjusting for age and cholesterol level since  $p \leq \alpha$ .

*There is evidence of an association between gender and risk of a coronary event after adjusting for age and cholesterol level.*

**The odds ratio for a coronary event is 12.44 for males versus females.**



## An Example: Multiple Logistic Regression

Use the regression model to **predict the risk of a coronary event for a 60 year old female with a cholesterol level of 150.**

$$\begin{aligned} p &= \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k}} \\ &= \frac{e^{-8.536 + 150 * 0.029 + 0 * 2.521 + 0.042 * 60}}{1 + e^{-8.536 + 150 * 0.029 + 0 * 2.521 + 0.042 * 60}} \\ &= \frac{e^{-1.66}}{1 + e^{-1.66}} = 15.9762\% \end{aligned}$$

*The risk of a coronary event for a 60 year old woman with a cholesterol level of 150 is 15.97%.*

Note in the above equation  **$x_M$  versus  $F = 0$**  since this is the dummy variable for males (which is equal to 0 for women).

## R commands: Generalized linear models (GLMs)

```
> glm(data$event~data$explanatory1 + data$explanatory2 + ... , family=
      binomial)
```

- ▶ In multiple logistic regression, use the `wald.test()` function (from `aod` package) to get p value for the global test (of all beta coefficients = 0)

## Multiple Logistic Regression

```
# multiple logistic regression
> data$male <- ifelse(data$sex == "M", 1, 0)
> m2 <- glm(data$event ~ data$chol + data$male + data$age, family=
  binomial)
> summary(m2)

# overall test
# install.package("aod")

> library(aod)
> wald.test(b=coef(m2), Sigma = vcov(m2), Terms = 2:4)

# Terms: An optional integer vector specifying which coefficients should
  be jointly tested
# Terms defines to compare which regression coefficients,
# here we want to compare the 2 to 4 (first is the intercept)
# It gives as a result Chi-Squared test results, and p-value of it
# if p is smaller than 0.05 you can reject the null hypothesis

# ORs per 1 unit increase
exp(cbind(OR = coef(m2), confint.default(m2)))
```